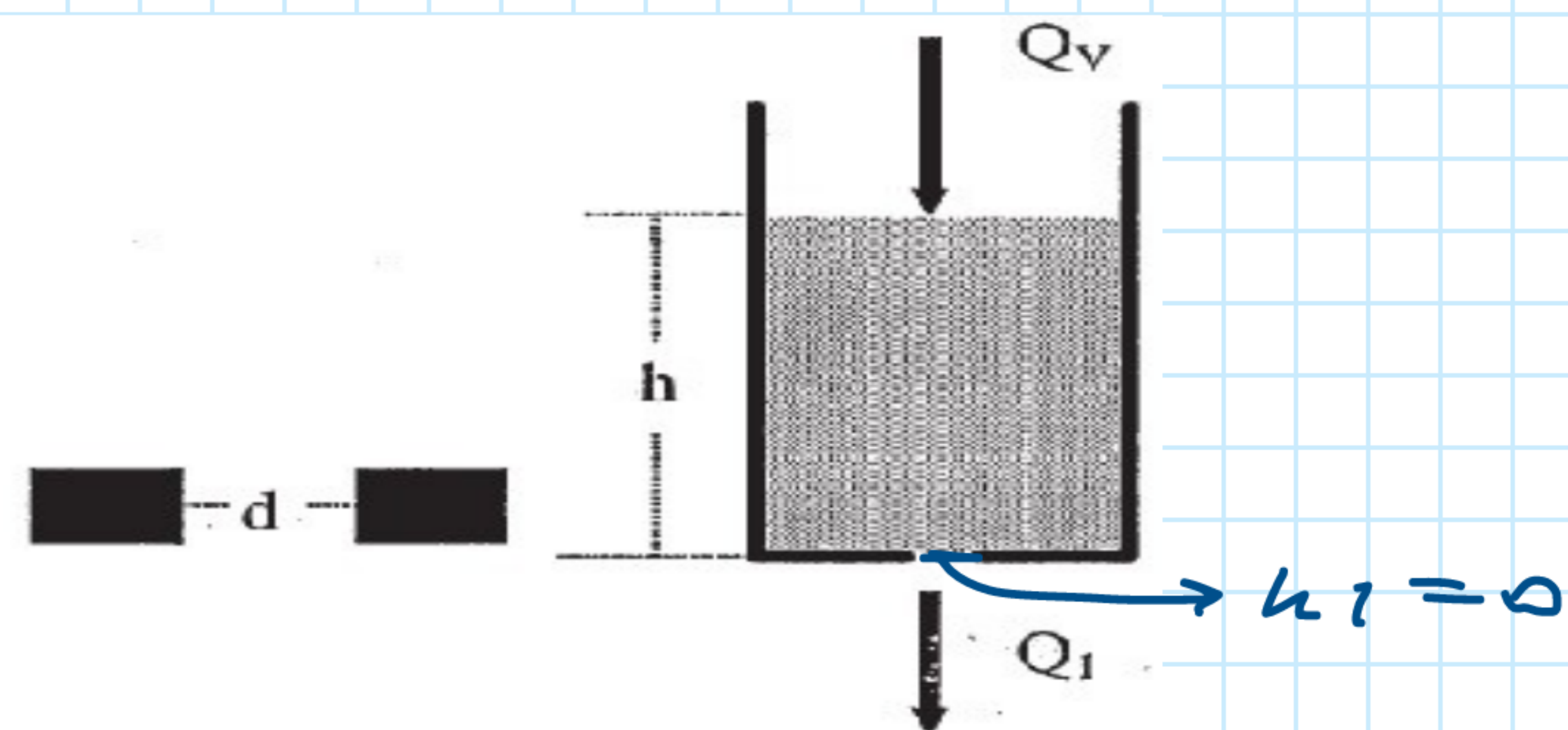


1.2.158 Në fundin e enës cilindrike, me seksion S që ka diametrin $D=0.5\text{ m}$, gjendet vrima rrethore S_1 me diametër $d=1\text{ cm}$. Të gjendet shpejtësia e rënies së nivelit të ujit në enë në çastin kur lartësia e ujit mbi vrimë është $h=0.2\text{ m}$?

Zgjidhje:



Të dhënat:

$$S =$$

$$D = 0.5\text{ m}$$

$$S_1 =$$

$$d = 1\text{ cm} = 1 \cdot 10^{-2}\text{ m}$$

$$h = 0.2\text{ m}$$

$$S = \pi r^2 = \frac{\pi \cdot D^2}{4}$$

$$r = \frac{D}{2}$$

$$S_1 = \pi r_1^2 = \frac{\pi d^2}{4}$$

$$r_1 = \frac{d}{2}$$

$$\left. \begin{array}{l} Q = S \cdot v \\ Q_1 = S_1 \cdot v_1 \end{array} \right\} S \cdot v = S_1 \cdot v_1 \Rightarrow v_1 = \frac{S}{S_1} v$$

$$Sgh + S \frac{v^2}{2} = Sgh_1 + S \frac{v_1^2}{2} \quad / : S$$

$$gh + \frac{v^2}{2} = \frac{v_1^2}{2}$$

$$gh + \frac{v^2}{2} = \frac{S^2}{S_1^2} \cdot \frac{v^2}{2} \quad / \cdot 2$$

$$2gh + v^2 = \frac{S^2}{S_1^2} \cdot v^2$$

$$2gh = \frac{S^2}{S_1^2} v^2 - v^2$$

$$2gh = \left(\frac{S^2}{S_1^2} - 1 \right) \cdot v^2 \quad / : \left(\frac{S^2}{S_1^2} - 1 \right)$$

$$\frac{2gh}{\frac{S^2}{S_1^2} - 1} = v^2 \quad / \sqrt{\quad}$$

$$v = \sqrt{\frac{2gh}{\frac{S^2}{S_1^2} - 1}} = \sqrt{\frac{2gh}{\left(\frac{\pi D^2}{4} \right)^2 - 1}}$$

$$v = \sqrt{\frac{2gh}{\frac{D^4}{d^4} - 1}} = \sqrt{\frac{2gh}{\frac{D^4 - d^4}{d^4}}}$$

$$v = d^2 \sqrt{\frac{2gh}{D^4 - d^4}} = (1 \cdot 10^{-2}\text{ m})^2 \cdot \sqrt{\frac{2 \cdot 9.81 \frac{\text{m}}{\text{s}^2} \cdot 0.2\text{ m}}{(0.5\text{ m})^4 - (1 \cdot 10^{-2}\text{ m})^4}}$$

$$v = 0.0792 \cdot 10^{-2} \frac{\text{m}}{\text{s}} \quad \text{D.Sh. të verifikohet rezultati}$$

$$\text{m}^2 \cdot \sqrt{\frac{\frac{\text{m}^2}{\text{s}^2}}{\frac{\text{m}^4}{1}}} = \text{m}^2 \sqrt{\frac{1}{\text{m}^2 \cdot \text{s}^2}} = \frac{\text{m}^2}{\text{m} \cdot \text{s}} = \frac{\text{m}}{\text{s}}$$